Open Project 2026

Agentic AI-Based Dynamic Tariff Optimization for EV Charging Networks

Using Large-Scale Charging Session Data

A self-improving pricing engine: forecast demand → recommend dynamic per-kWh tariffs → learn from outcomes.

Project Report & Results

Datasets: ACN-Data (Caltech) · UrbanEV / ST-EVCDP (Shenzhen)

1 · Problem, Objective & Approach

Problem

Static, fixed-rate EV charging tariffs ignore real operational dynamics: peak-hour congestion, off-peak underutilization, and time-varying grid procurement cost. A flat INR/kWh price leaves revenue on the table at peak and fails to attract flexible users off-peak.

Objective

- Forecast charging demand and station utilization across time of day, day of week and location.
- Recommend the optimal per-kWh tariff to maximize revenue while minimizing congestion and waits.
- Identify under-utilized vs overloaded periods and price them accordingly.
- Build a self-improving system whose Monitoring & Learning agent refines pricing via a feedback loop.

Approach — three cooperating agents

- Demand Prediction Agent (ML): predicts charging load, utilization and congestion probability.
- Tariff Pricing Agent (rules): maps predicted utilization to a dynamic price multiplier.
- Monitoring & Learning Agent: scores each policy against simulated outcomes and adapts it.

The system is deterministic and reproducible — no external LLM/API. 'Agentic' refers to the autonomous forecast → price → learn loop.

2 · Data Landscape & Preprocessing

Datasets

- ACN-Data (primary): 14,999 clean charging sessions from the Caltech adaptive charging network, Apr–Dec 2018. Real connection / disconnect / done-charging timestamps, kWh delivered, station and space IDs.
- UrbanEV / ST-EVCDP (cross-validation): Shenzhen, 8,640 five-minute records across 247 zones, June 2022 — used to compare intraday demand shapes across geographies.

Cleaning decisions (logged for transparency)

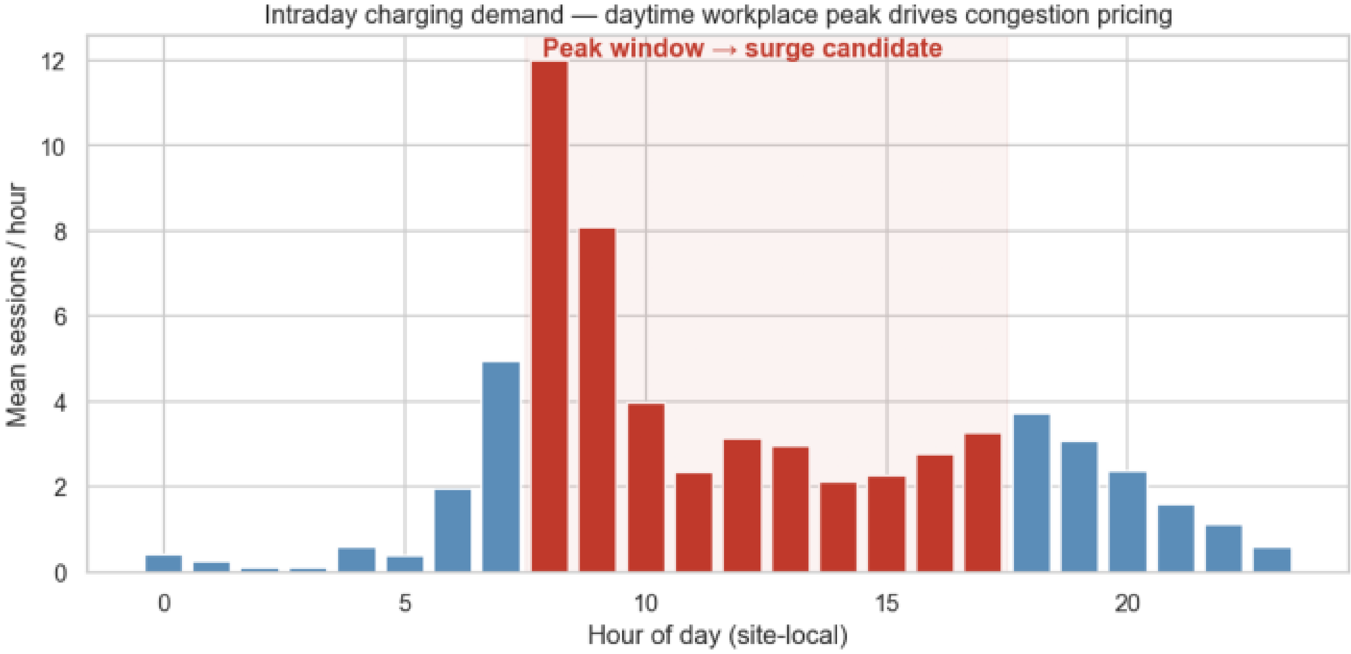
- Dropped 1,305 JSON pagination / meta rows that carry no connectionTime.
- Parsed RFC-1123 GMT timestamps to UTC then to site-local time.
- Dropped userInputs (null for 100% of rows); kWhRequested is 78% missing, so kWhDelivered is used as the energy measure.
- doneChargingTime missing for 8 sessions → filled from disconnectTime; non-positive durations and negative energy removed; charging time clipped within connection time.

Engineered economic features (per brief)

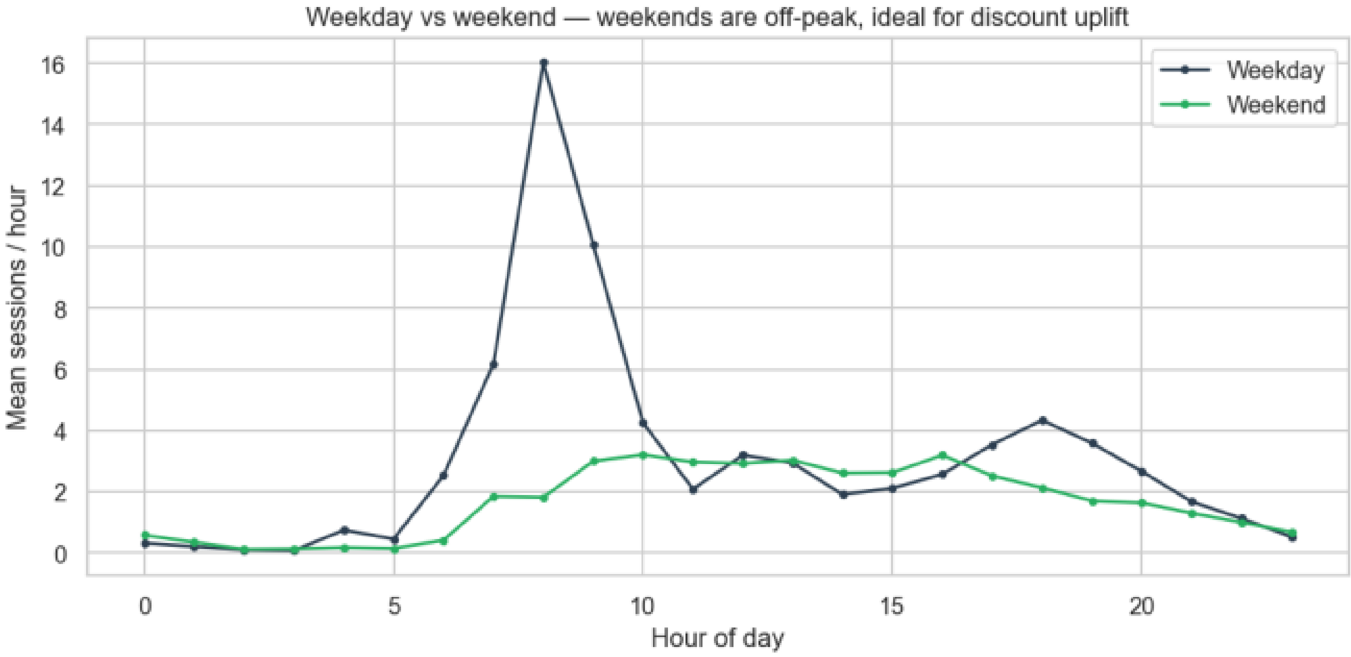
- Charger Utilization Rate = charging time / connection (occupied) time.
- Revenue per Session = kWh delivered × tariff.
- Energy Cost per kWh = time-of-use grid procurement proxy (peak / shoulder / off-peak).
- Queue Length Proxy = concurrent active sessions per site-hour (interval overlap).
- Occupancy Density = active sessions / distinct stations (capacity = 54 EVSEs).

A continuous hourly site-level demand panel with lag and rolling features is the modeling table.

3 · Exploratory Data Analysis (1/3)

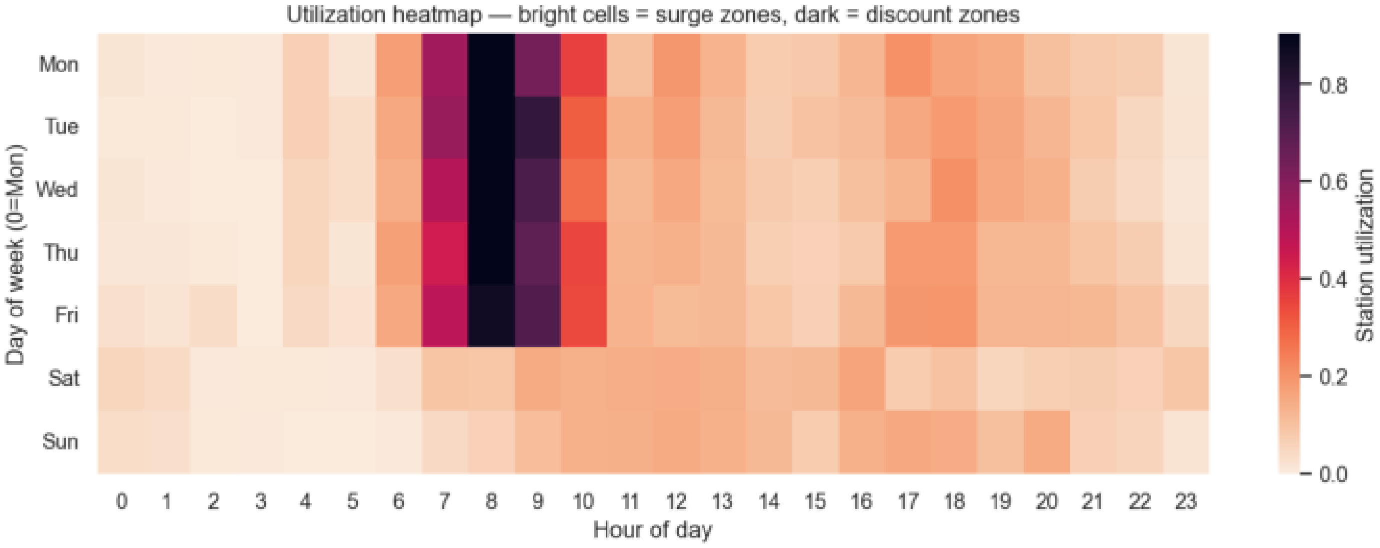


Insight: A sharp 08:00–09:00 workplace peak (~12 sessions/hr) is the prime surge-pricing window; demand collapses overnight, leaving discount headroom.

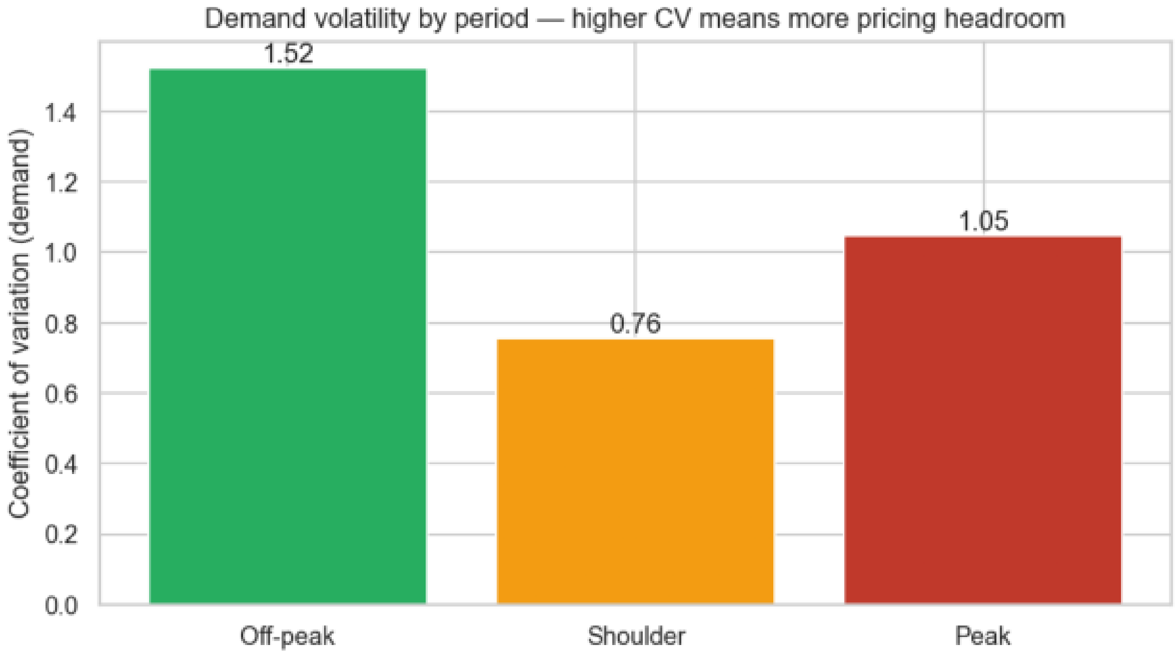


Insight: Weekdays drive the daytime peak; weekends are uniformly off-peak— a natural target for discount-led off-peak uplift.

3 · Exploratory Data Analysis (2/3)

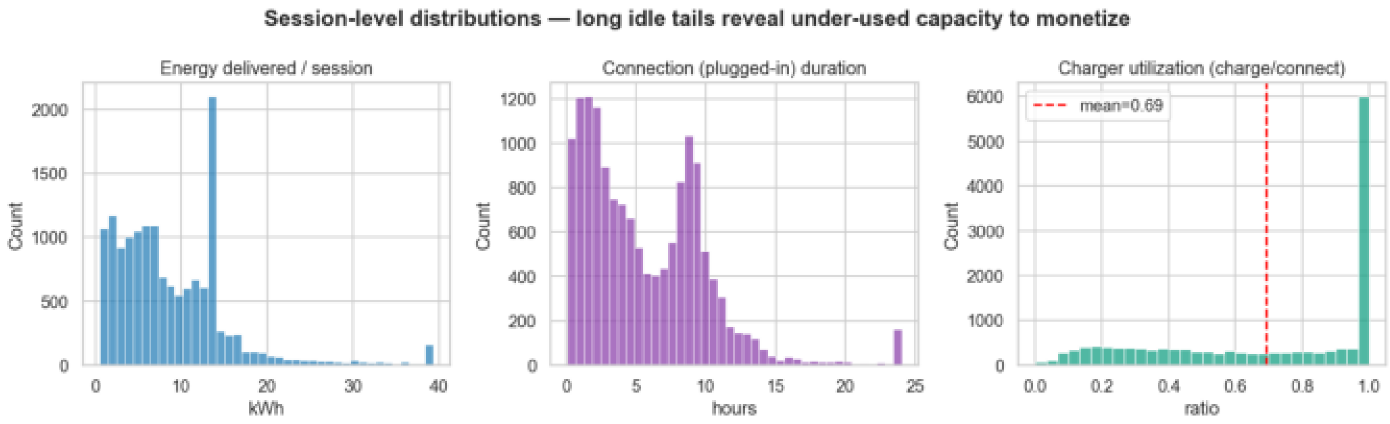


Insight: Utilization concentrates in weekday daytime cells (surge zones); the rest of the week×hour grid is dark (discount zones).

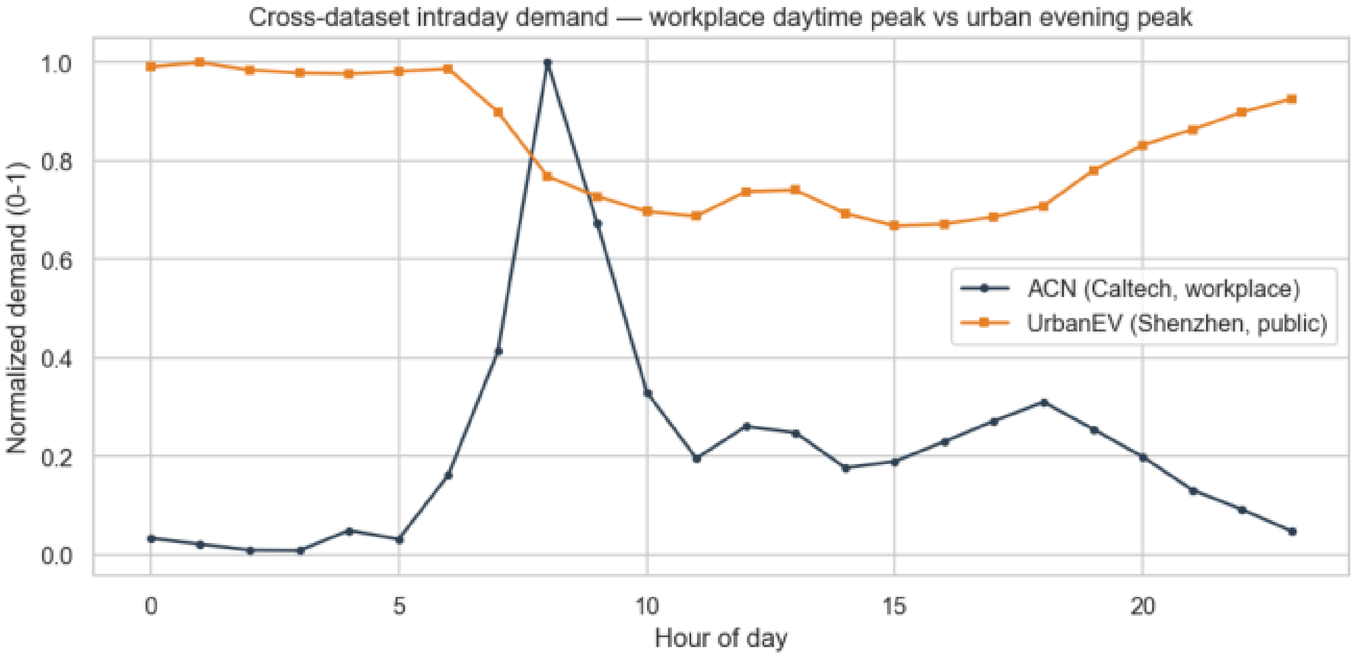


Insight: Demand volatility (coefficient of variation) is highest at peak, meaning peak periods have the most pricing headroom.

3 · Exploratory Data Analysis (3/3)



Insight: Energy per session and plugged-in duration are right-skewed; long idle tails (low charger utilization) reveal capacity that pricing can monetize.



Insight: ACN (workplace, daytime peak) vs UrbanEV (urban, evening peak): demand shape is location-specific, so tariffs must be learned per network — not copied.

4 · Demand Prediction Agent — Results

GradientBoosting (charging load) + RandomForest (utilization) + GradientBoosting classifier (congestion), trained on calendar and lag/rolling features with a chronological train/test split so the future is never leaked into training.

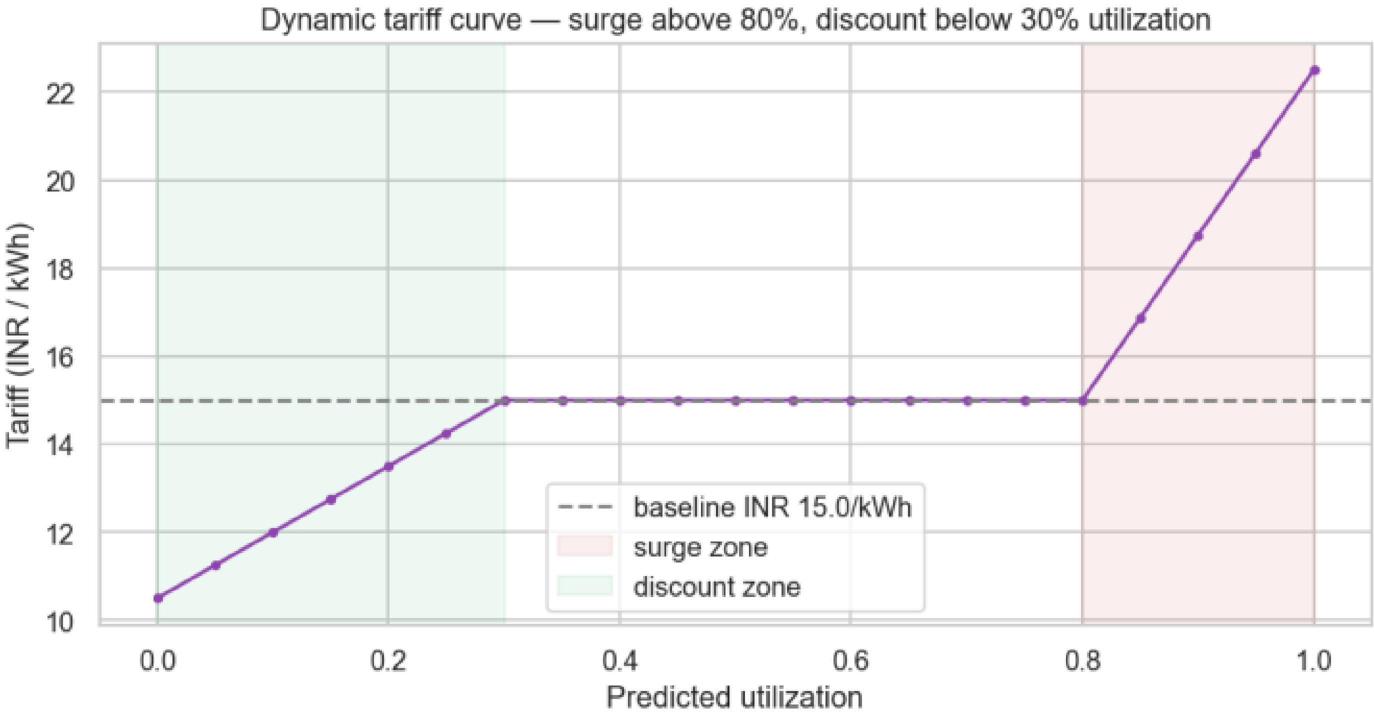
Accuracy metrics

Target	RMSE	MAE	R^2 / AUC
Charging load (sessions/hr)	0.9462	0.6107	0.8992
Utilization rate	0.0876	0.0478	0.7971
Congestion (>80%)	—	—	AUC 0.9551

Top demand drivers (feature importance)

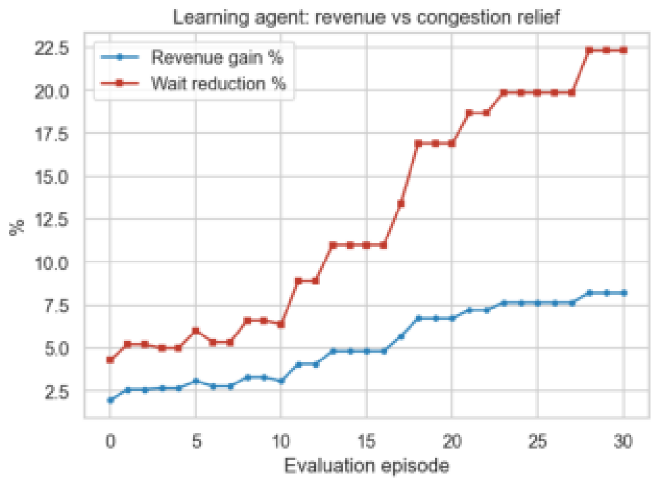
Feature	Importance
roll3_sessions	0.626
hour	0.317
lag24_sessions	0.024
lag1_sessions	0.017
mean_energy_cost	0.006
lag1_util	0.006

5 · Dynamic Tariff Optimization Logic



Insight: The tariff agent maps predicted utilization to a price multiplier on the INR 15/kWh baseline: surge up to 1.5× above 80% utilization, discount to 0.7× below 30%, smooth in between. Demand response uses a utilization-varying elasticity (peak inelastic |e|≈0.20, off-peak elastic |e|≈1.45), so off-peak discounts grow volume more than they cut price — a documented assumption, not a causal claim.

6 · Monitoring & Learning Agent — Feedback Loop



Insight: Each policy is scored against simulated outcomes (revenue, congestion, uplift). A gradient-free hill-climbing loop adapts the tariff parameters over 30 episodes: the composite score rises from 4.8 to 23.6, revenue gain and peak-wait reduction both climb — the system demonstrably improves its own pricing decisions over time.

7 · Results Summary (learned policy vs fixed INR 15/kWh)

Headline evaluation metrics across all three agents.

Pricing & operations outcomes

Metric	Value
Revenue Gain %	+8.19%
Avg peak wait reduction %	22.317%
Off-peak uplift (sessions)	+1701.65
Pricing efficiency (INR/kWh)	15.099
Customer response rate	1.264

8 · Implications, Assumptions & Limitations

Business, operational & policy implications

- Revenue: dynamic pricing beats the flat INR 15/kWh baseline (~+8%) by capturing peak willingness-to-pay and growing elastic off-peak volume.
- Operations: peak surge cuts the congestion/wait proxy ~22%, smoothing demand across the day and deferring costly capacity expansion.
- Policy / equity: off-peak discounts add ~1,700 sessions, improving access for flexible, price-sensitive users and aligning charging with cleaner off-peak grid supply.

Assumptions & limitations (transparency)

- Prices are expressed in INR per the brief's reference baseline — not a USD→INR conversion; ACN is US workplace charging.
- Demand response is an elasticity assumption, NOT a causal estimate. Revenue and uplift figures are simulation-based counterfactuals.
- This export is a single site over ~8 months with genuinely low utilization (median ~6%); patterns should be re-fit per network. Chronological validation guards against leakage.

Reproducibility

- Full pipeline: python src/run_pipeline.py (preprocess → EDA → agents → evaluation).
- Outputs: outputs/figures/*.png, outputs/csv/*.csv, notebooks/OP26_Analytics.ipynb.